# A Simple Introduction to Monte Carlo Simulations for a priori Power Analyses

# Outline

- Importance of Sample size
- ‘Classic’ Power
- True Positive Rate
- Monte Carlo Simulation
- Example in R
- Applications

# Importance of Sample Size

- N too small
  - Power too low to detect effect
- N too large
  - Costs and resources
  - Statistical Significance: ‘The Null is never true’
- Solution: a priori minimum sample size determination

# The ‘classic’ concept of power

- Test: Correct positive decision
- There is a true effect, and we are able to detect it



# The ‘classic’ concept of power

- ‘classic’ power is possible to calculate
  - Two-sample t-test, paired t-test, …
- However, some analyses do NOT have analytical solutions
- Monte Carlo simulations can help!

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- … given certain probability distributions (such as H1)!

# Monte Carlo Simulation for TPR

- How does it work?

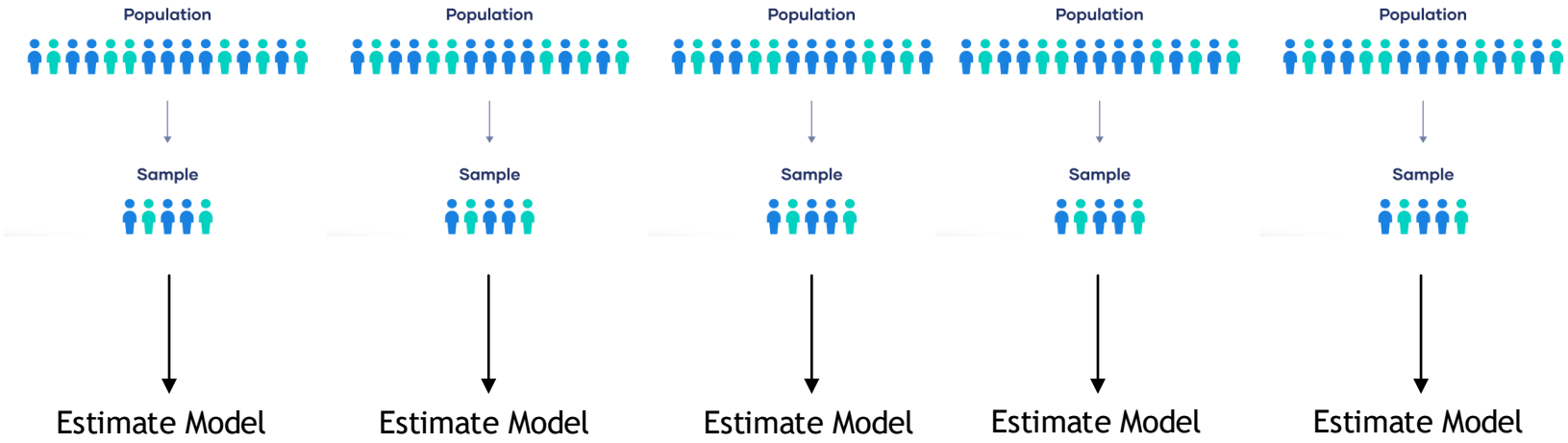
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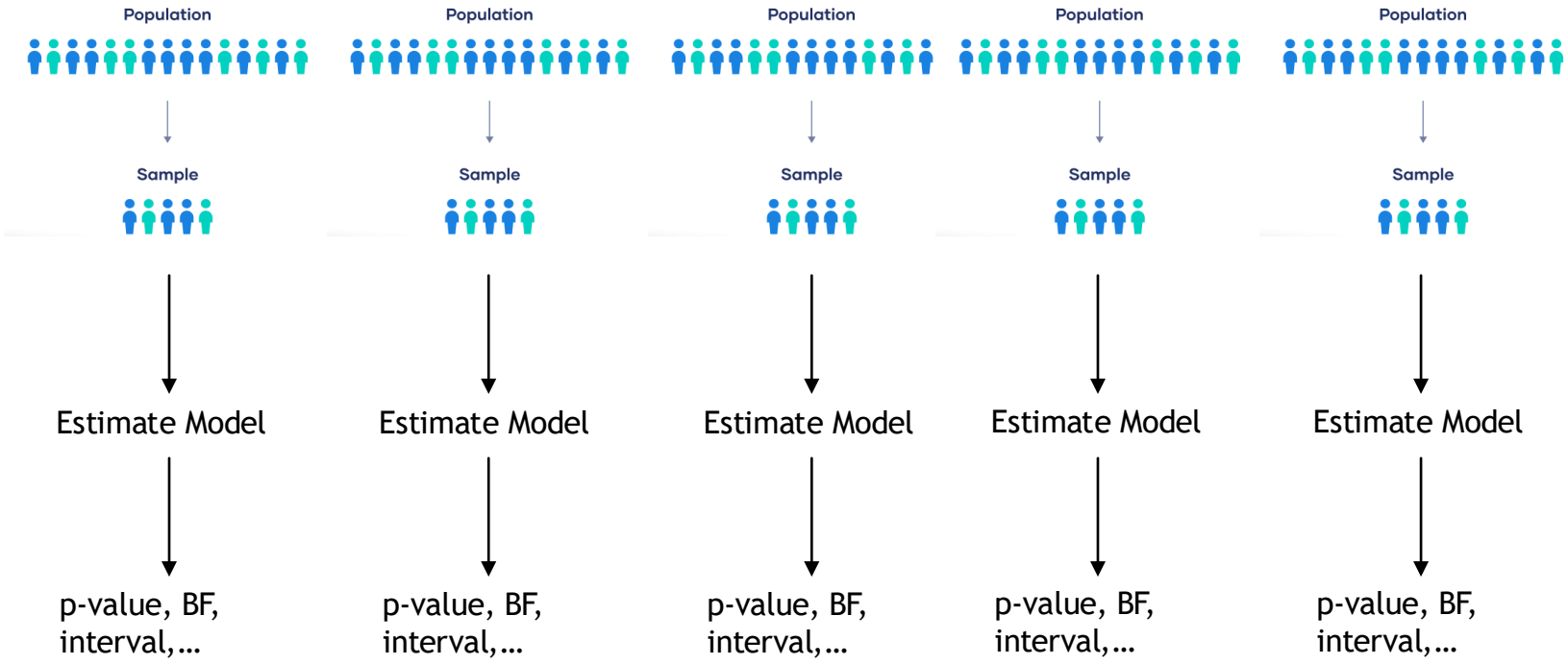
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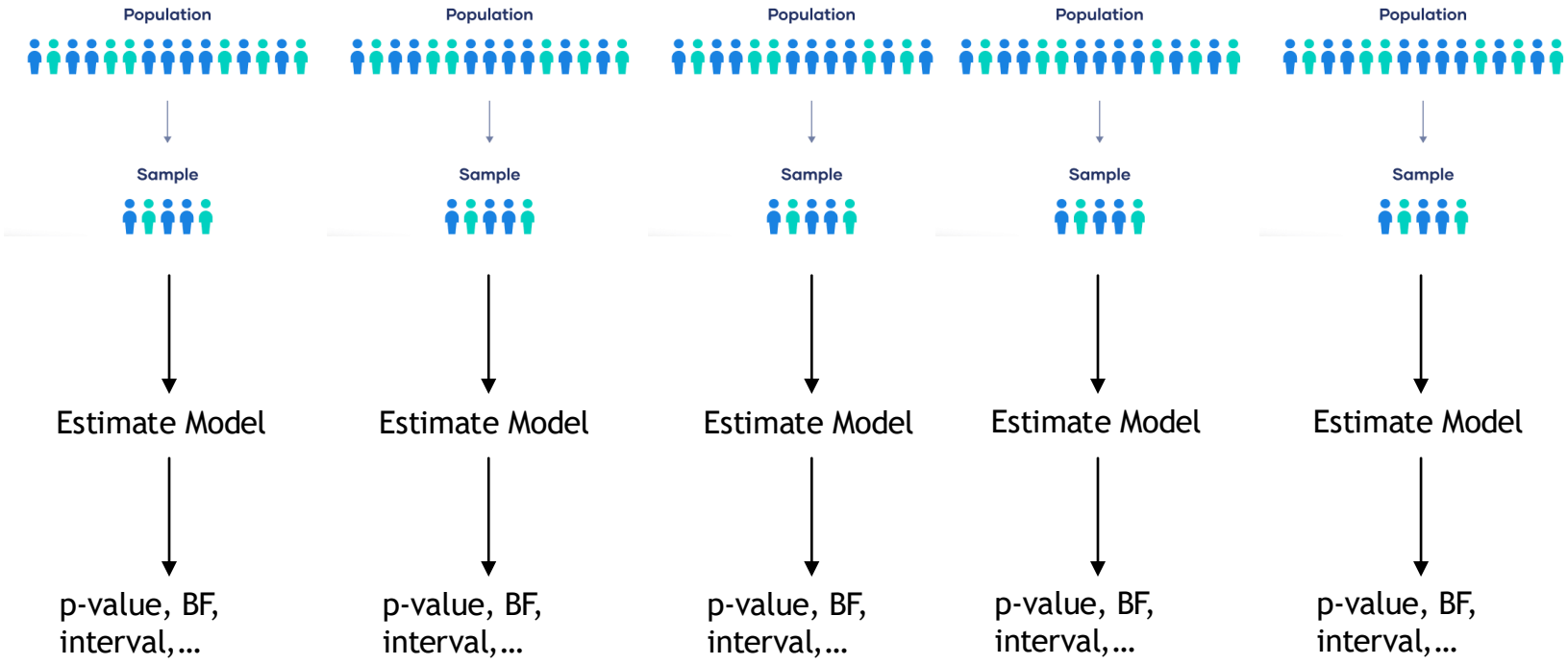
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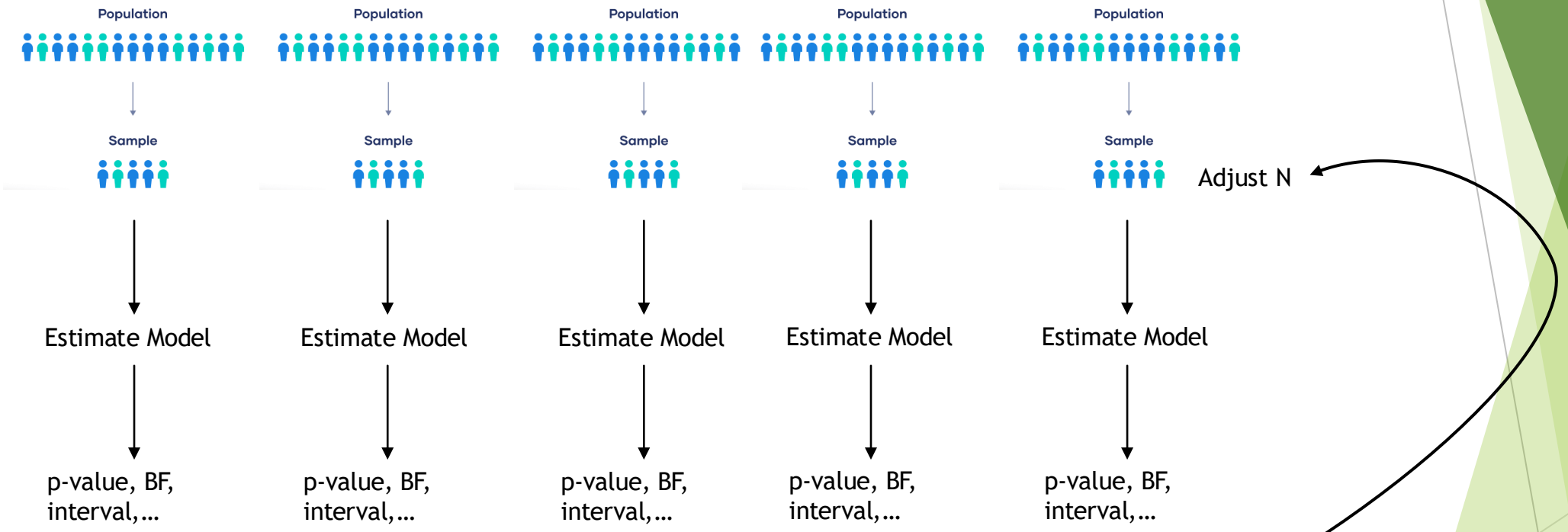


# Monte Carlo Simulation for TPR




Calculate the ‘power’ -> TPR = true positives / iterations

# Sample Size Determination




Calculate the ‘power’ -> TPR = true positives / iterations

# Example in R:
# One-way ANOVA with 4 Groups

- Is there a difference in grades of students from Munich, Berlin, Hamburg, Cologne?
- ANOVA assumptions
  - Normality within groups
  - Homoscedasticity within groups
  - Independence

# Applications

- What sample size do we need for a TPR > 0.80 of…
  - obtaining BF > 10 to establigh evidence for a Hypothesis?
  - obtaining a significant indirect effect in a mediation model?
  - obtaining a confidence interval which is within a region of practical equivalence?

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- Research assisstantship
  - Sample size determination for TPR of bayesian two-way ANOVA
- Traineeship
  - Influence of small sample sizes on the correct partitionings (TPR) of items into measurement scales

# Discussion

- Questions?
- Thoughts?